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Implications of a combined biostratigraphic and palaeomagnetic study of the Umbrian Maiolica Formation

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The study of a 275.5 m thick section of white, pelagic limestones occupying the valley of the Fonte del Giordano river on the southern slope of Mt. Montagnola has yielded a biostratigraphically controlled clear magnetic reversal pattern after thermal cleaning. The magnetic stratigraphy of the lower 131 m of the section (Calpionellid zones) is correlatable with the M-sequence of oceanic magnetic anomalies between M-19 and M-14. The reversal stratigraphy of the upper 81.5 m of the section (Radiolaria zone) has also been tied to the oceanic polarity time scale by making linear interpolation for a missing 63 m thickness underneath it.

Besides the Fonte del Giordano section two Berriasian outcrops each with a different bedding attitude were studied at Gubbio and near Cagli for tectonic tilt test giving positive results. The mean palaeomagnetic pole position for the Late Jurassic/Early Cretaceous after bedding correction is: $\Phi = 19.1°$, $\Lambda = 288.2°$, $k = 148.7$, $\alpha_{95} = 10.2°$ ($N = 3$), confirming the presence of a large swing in the polar path, a common behaviour of apparent polar wandering for the peri-Adriatic region during this time.

1. Introduction

Márton and Márton [1] have proposed that peninsular Italy, Istria and the Transdanubian Central Mountains may be characterized by a common apparent polar wander curve for the Mesozoic implying that these units were not in relative motion during the whole Mesozoic. They have shown that after correcting for a post-Mesozoic counterclockwise rotation between the above units and Africa [2–4] the common apparent polar wander curve agrees closely with the polar pattern obtained from direct palaeomagnetic determinations for Africa and South America [5].

The Late Jurassic/Early Cretaceous segment of the rotated common apparent polar wander curve is more elongated in the southeastern direction than that of the inferred African polar wander curve [3] and the palaeomagnetic results pertaining to this period of time can be fitted either into the true African polar pattern [1,4,6] or into the inferred one [2,7,8].

It is obvious that more palaeomagnetic data are needed especially from Italy and Istria in order to substantiate the validity of a common apparent polar wander curve for this particular time. In Italy, the pelagic Maiolica limestones of Tithonian/Neocomian age have suggested for a study in this direction (B. D'Argenio, personal communication) and these rocks were sampled in Umbria at outcrops shown to us by Professor G. Pialli.

2. Sampling localities

The miogeosynclinal sediments of the Umbria-Marche region consist of several lithological se-

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quences ranging from Triassic to late Miocene in age. Sampling was concentrated mainly on the white, micritic limestones with chert beds and nodules known as the Maiolica Formation. In this region, the Maiolica is of the order of 300 m thick and contains a rich fauna of Tithonian and Neocomian calpionellids [9].

The main sampled section is located on the southern slope of Mt. Montagnola near the village of Cerreto and occupies the valley of the Fonte del Giordano river (Fig. 1). The sampled sequence of the Fonte del Giordano section includes the uppermost beds of the Calcari Diasprini of early Tithonian age and the overlying Maiolica limestones for about 270 m upwards. In the lower 124 m of this interval the age has been established on the basis of calpionellids as late Tithonian through Valanginian. Higher up in the section the limestones lack calpionellids and the faunal assemblage (Radiolaria and Foraminifera) suggests an Hauterivian age.

The total stratigraphic thickness sampled in the Fonte del Giordano section is 275.5 m excluding two intervals between 18 and 30 m and between 131 and 194 m which could not be sampled.

Due to expected variations in the sedimentation rate over the different parts of the section (G. Pialli, personal communication) the following (average) sampling distances were adopted: 0.7 m from 0 to 18 m, 2 m from 30 to 131 m and 2.7 m from 194 to 275.5 m.

Two other outcrops of Maiolica limestone were sampled near Gubbio and Cagli respectively (Fig. 1), in which the bedding attitudes are different both from each other and also from those in the Fonte del Giordano section. Eleven, respectively 8 samples were collected at a rate of 1 sample per 2 m and on the basis of calpionellid content they all are of Berriasian age.

From all sections magnetically oriented hand samples were collected which were subsequently drilled in the Bologna palaeomagnetic laboratory.

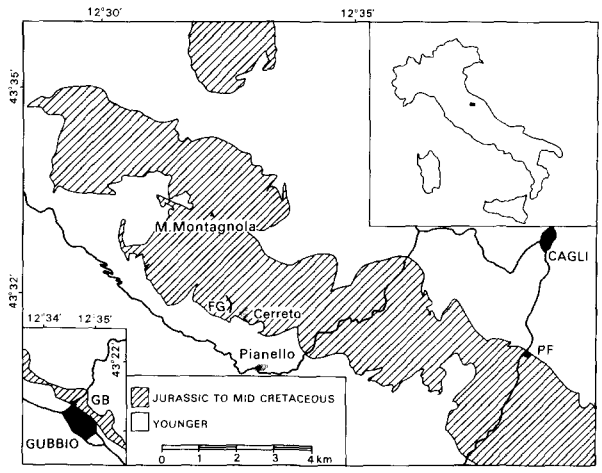


Fig. 1. Location of the Fonte del Giordano (FG), Gubbio (GB) and Cagli (PF) sections.

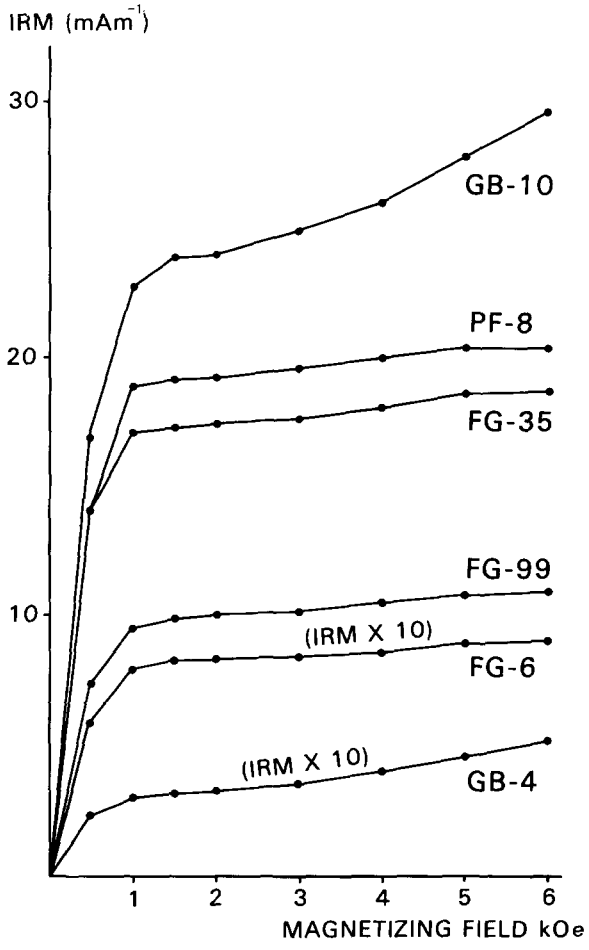


Fig. 2. IRM acquisition behaviour of samples from the FG, GB and PF sections.

3. Magnetic properties

The remanence measurements were carried out in Bologna and Budapest using JR-4 spinner magnetometers with an r.m.s. noise level of 4×10^{-6} A/m. The treatment of the samples depended on the results of IRM studies, a technique which has proved very useful particularly with marine limestones [10].

The observations of IRM (isothermal remanent magnetization) acquisition in increasing magnetic fields up to 6 kOe (Fig. 2) suggest that one of the carriers of the IRM in these limestones is a low (0–1 kOe) coercivity remanence fraction probably magnetite. The increase of the IRM in the range of 1–6 kOe can be attributed to the presence of haematite whose contribution to the IRM is quite small except for the samples from Gubbio (GB-4, GB-10) in which the significance of the haematite is comparable to that of the magnetite.

Fig. 3 shows examples of the thermal variation of intensity of a composite IRM first induced in the samples in a 6 kOe field in one direction then in a 1.5 kOe field in the opposite direction both at room temperature before heating. The steep decrease of this composite IRM between 400 and 540 °C can be attributed to the unblocking of the magnetite remanence. Further heating reveals in all studied samples that part of the low (< 1.5 kOe) coercivity remanence fraction only unblocks above the Curie temperature of the magnetite but completely disappears at 650 °C. This IRM component is attributed to a soft ($h_{cr} < 1.5$ kOe) ferromagnetic haematite fraction (specularite?), whereas the one surviving 650 °C is carried by a hard (1.5 kOe $< h_{cr} < 6$ kOe) antiferromagnetic haematite phase [21] as indicated by the negative deflection toward the end of four of the temperature vs. IRM curves (Fig. 3).

One of the samples from the Gubbio locality (GB-10 in Fig. 3) displays an increase of IRM between room temperature and 120 °C due to the loss of a high (1.5 kOe $< h_{cr} < 6$ kOe) coercivity remanence fraction of low unblocking temperature

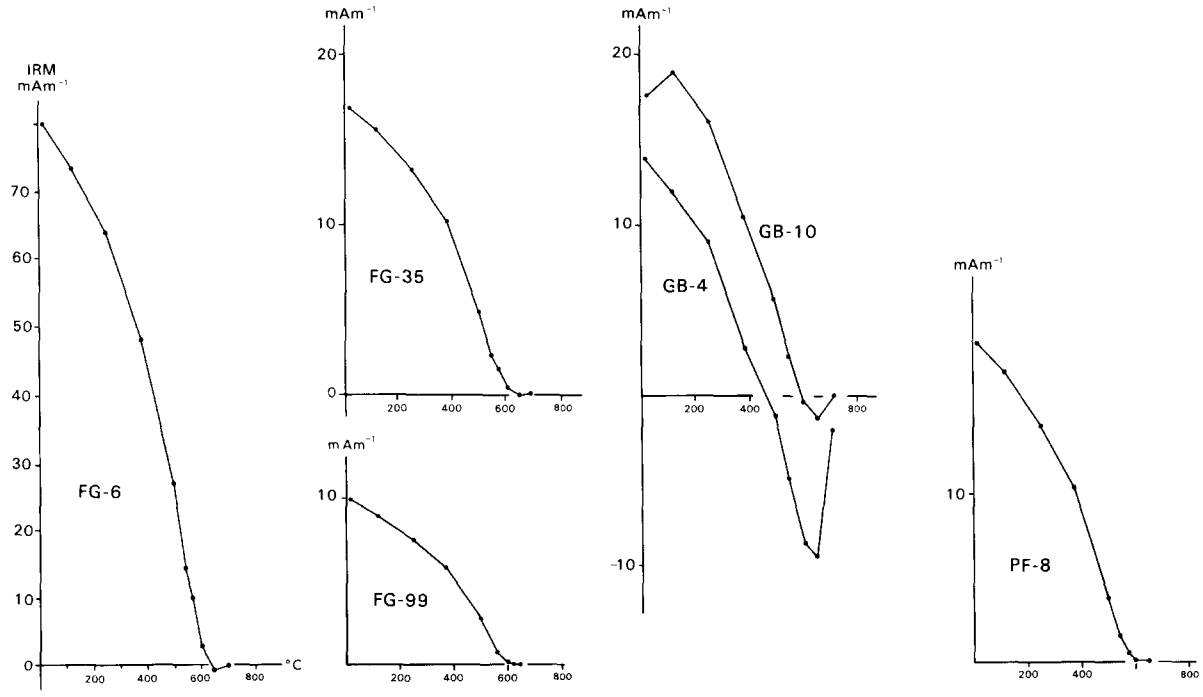


Fig. 3. Demagnetization curves showing the thermal variation of intensity of a composite IRM first induced in a 6 kOe field in one direction, then in a 1.5 kOe field in the opposite direction at room temperature.

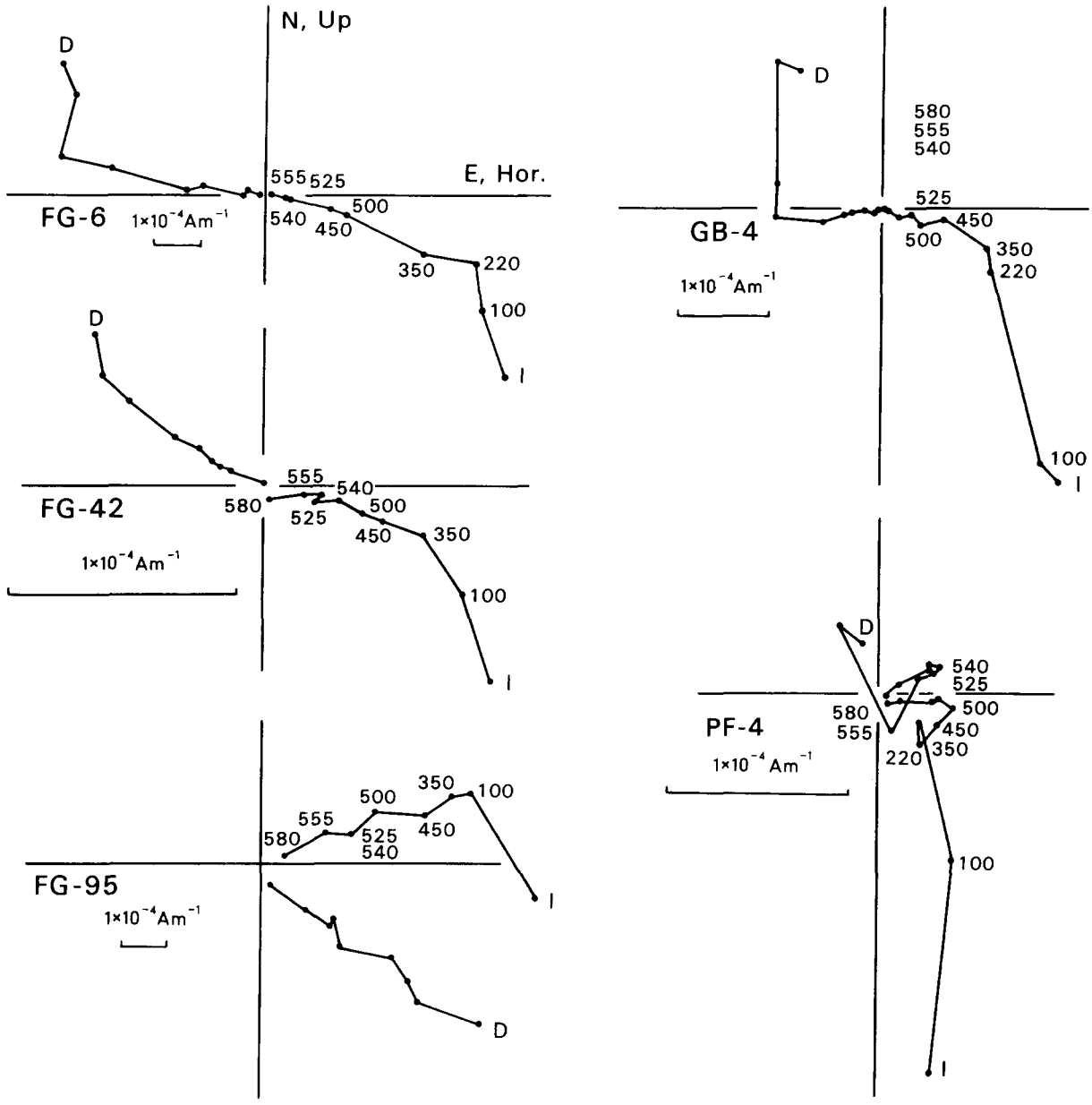


Fig. 4. Variation of the declination (D) and inclination (I) of NRM during progressive thermal demagnetization.

which may be identified with goethite.

Because of the presence of haematite and the occasional occurrence of goethite besides the magnetite thermal demagnetization was chosen for magnetic cleaning. In order to monitor chemical changes during heating the magnetic susceptibility, κ, was measured after each heating step using a KLY-1 low-field susceptibility bridge.

Some 30 pilot specimens were totally demagnetized stepwise using Schonstedt TSD-1 ovens. Respective thermal demagnetization diagrams are reproduced in Fig. 4. Major susceptibility changes

during heating were recorded only for a single pilot specimen (GB-4 in Fig. 4), however, they do not seem to have influenced the demagnetization behaviour of the natural remanence.

Generally (Fig. 4), a dominant remanence component could be isolated after demagnetization between 500 °C (occasionally 450) and 555 °C which is assumed to be the characteristic magnetization carried by the magnetite whose presence has been shown by the IRM experiments. Reliability indicators [11] allow to assume further that the characteristic magnetization was formed shortly after deposition so that the time lag between deposition and acquisition of remanence was negligibly short relative to the duration of a polarity state of the geomagnetic field.

4. Palaeomagnetic results and their implication

Two to four specimens were demagnetized from each sample in one to four steps between 450 °C (occasionally 420) and 540 °C and the direction of the characteristic remanence was defined for a sample by averaging the specimen directions. The remanence directions for specimens from the same sample measured separately in the Bologna and Budapest laboratories usually agreed within a few degrees. Significant differences were only noticed with a few weak specimens of intensities near the noise level of the magnetometers.

The average magnetic intensity for all three localities is of the order of 5×10^{-5} A/m after demagnetization. In the Fonte del Giordano section, the intensity increases almost an order of magnitude relative to the average towards the base while it is about half the average over the whole upper part. These intensity changes may be indicative of the variations in the sedimentation rate, the smaller intensities being due to faster deposition and vice versa (see section 6).

From the systematic variation of the bedding attitude along the section the Fonte del Giordano sequence has been recognized to constitute a local structure of plunging asymmetrical syncline. The normal to the best fitting plane to the bedding poles which defines the plunge of the fold is (27.8°, 153.8°). It has also been found that after correction for the plunge the bedding poles could be grouped into four different clusters corresponding to the lowermost (0–4.5 m), lower (5–18 m), middle (30–131 m) and upper (194–275.5 m) part of the section. Taking the mean for each of the clusters and combining each mean with the plunge four rotations, one for each subsection, were derived for total tectonic correction and they are as follows: (8.4°, 27.1°, 27°), (0.8°, 60.5°, 27°), (−3.4°, 77.6°. 27.9°) and (−6.6°, 91.7°, 31°), respectively. Here, the first two numbers in the brackets define the dip and azimuth of the rotation axis in the local geomagnetic system of coordinates and the third gives the amount of counterclockwise rotation.

In the Fonte del Giordano section the directions of the characteristic remanence were tectonically corrected using the above rotations while in the Gubbio and Cagli sections where the bedding attitudes are monoclinal simple tilt corrections were applied.

All relevant palaeomagnetic data derived from this study are presented in Table 1. The Fonte del Giordano and the Gubbio and Cagli directions respond positively to the tectonic fold test indicating that the palaeomagnetic directions predate the tectonic folding.

The new Umbrian palaeomagnetic pole for the Late Jurassic/Early Cretaceous is shown in Fig. 5 together with other relevant palaeomagnetic poles for comparison. The Neocomian and Kimmeridgian poles [2] as well as the Barremian pole [6] are also from Umbria. From the stable Adriatic foreland Gargano is represented by an Early Cretaceous pole [8] and Istria by two [7]. The Tithonian and the Neocomian poles are averages obtained from four, respectively three geographically distributed localities in the Transdanubian Central Mountains [1]. The agreement of the Istria and Gargano poles is excellent and they also are close to the Umbria poles of VandenBerg et al. [2].

On the other hand, the present pole position fits to the Barremian pole for Umbria [6] and both are displaced south-southeast from the former group of poles, i.e., in the direction where the Transdanubian poles for the Late Jurassic/Early Cretaceous are situated [1]. Lowrie et al. [6] have shown that the Barremian/Aptian poles for the southern

TABLE 1

Palaeomagnetic results from Umbria

No.	Locality	Number of N/R samples	Before tectonic correction				After tectonic correction				Palaeomagnetic pole			
			D	I	k	α_{95}	D	I	k	α_{95}	lat.	long.	Δp	Δm
1.	Fonte del Giordano (FG) $\varphi = 43.53°, \lambda = 12.55°$	63/39	291.5	15.9	46.5	13.6	281.7	33.5	835.7	3.2	20.9	288.4	2.1	3.6
2.	Gubbio (GB) $\varphi = 43.33°, \lambda = 12.57°$	9/2	267.1	16.5	13.1	13.1	273.2	32.3	13.1	13.1	14.2	293.4	8.3	14.8
3.	Cagli $\varphi = 43.52°, \lambda = 12.65°$	3/4	260.7	8.0	20.4	13.7	286.3	27.8	20.4	13.7	21.9	282.5	8.2	15.0
													k	α_{95}
	Mean (1+2+3):		272.9	13.8	24.3	25.5	280.5	31.3	158.3	9.8	19.1	288.2	148.7	10.1

N = number of normal samples; R = number of reversed samples; D = palaeomagnetic declination (°); I = palaeomagnetic inclination (°); lat. = palaeomagnetic pole latitude (°); long. = palaeomagnetic pole longitude (°); k, α_{95}, Δp, Δm = statistical parameters [12].

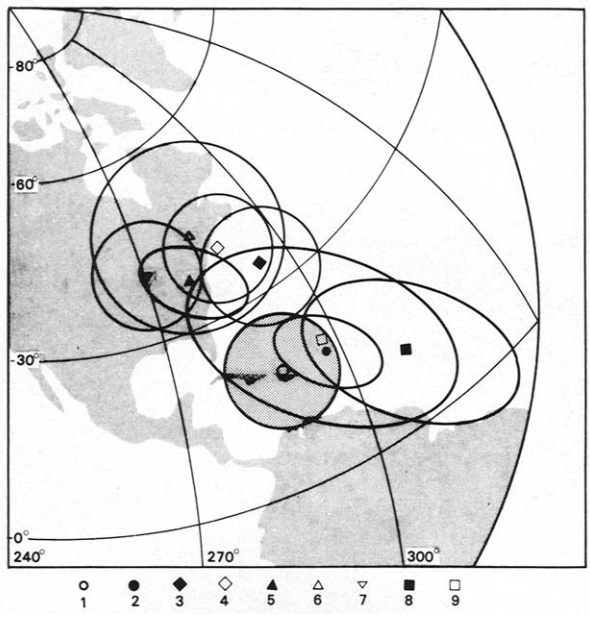


Fig. 5. Comparison of Late Jurassic/Early Cretaceous palaeomagnetic poles from the Mediterranean region. The poles are given with their 95% confidence circles or ovals and they are as follows: *1* = Umbria (present study); *2* = Umbria, Barremian [6]; *3* = Umbria, Kimmeridgian [2]; *4* = Umbria, Neocomian [2]; *5* = Gargano, Early Cretaceous [8]; *6* = Istria, Valanginian/Hauterivian [7]; *7* = Istria, Barremian/Aptian [7]; *8* = Transdanubian Central Mountains, Tithonian [1]; *9* = Transdanubian Central Mountains, Neocomian [1].

Alps are also moved southeast with respect to the Late Cretaceous poles, the Barremian pole being the most divergent.

On the basis of the latter group of observations, it has been suggested [1] that a large apparent polar wandering during Late Jurassic/Early Cretaceous time is common to all Gondwana fragments including those which today compose the peri-Adriatic region. In the light of this assumption the Late Jurassic/Early Cretaceous is also a time of fast apparent polar wandering. Consequently, any dating error in the age of the palaeomagnetically studied rocks or in that of the magnetization or in both will have a more serious effect on the respective pole position than it would have during intervals of time of slow polar movement.

All results enumerated above have been derived from the palaeomagnetism of pelagic limestones which usually have a complex magnetic history [10]. Though all studies have used efficient cleaning techniques to unravel the characteristic remanence the latter may not always be contemporaneous with the stratigraphical age of the rock. It is definitely not coeval with the rock if, e.g., only normal samples are measured for a mixed polarity period of time. Another difficulty is that some of these rocks are characterized by extremely weak remanences. On demagnetization, the intensity may drop to or below the noise level of the measuring instrument before perfect cleaning is attained and imperfectly cleaned remanences may result in biassed pole positions.

Tectonic effects, if not properly evaluated and corrected for, are also liable to introduce errors in the palaeomagnetic declination but these are serious only for steeply dipping fold axes. Small dips (tilts) are more dangerous because an insignificant improvement of the grouping of the magnetic directions can be interpreted mistakenly as proof for the ancient origin of the respective magnetization.

It is not our task, however, to judge to what extent the pole positions in Fig. 5 are influenced by the above described dating and interpretation errors. For the time being we are content to observe that the present palaeomagnetic result for the Tithonian through Hauterivian stages from Umbria is in good agreement with a southeasterly swing in the peri-Adriatic apparent polar wander curve during the Late Jurassic/Early Cretaceous [1] which was originally inferred from contemporaneous palaeomagnetic data from Umbria and the southern Alps [6] and the Transdanubian Central Mountains [1].

5. Biostratigraphy

The biostratigraphic subdivision of the sampled pelagic sequences has been based on microfaunal assemblages. The number of thin sections studied equals that of the palaeomagnetic samples.

In the Fonte del Giordano section five biozones were recognized ranging from the Tithonian through Hauterivian (Fig. 6).

The lowermost 6 m of the section can be referred to as the *Saccocoma* Zone (Early Tithonian)

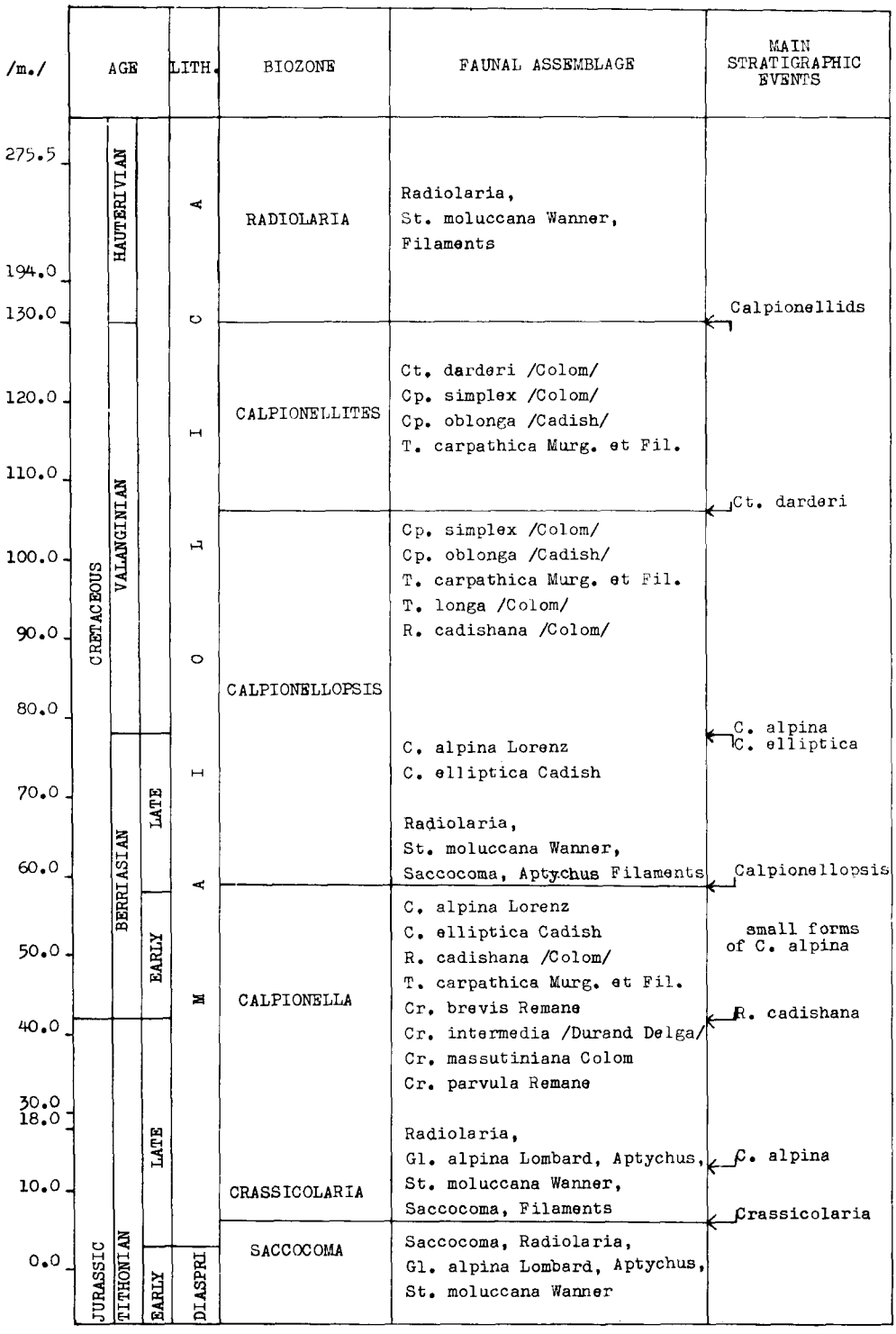


Fig. 6. Biostratigraphical zonation of the Fonte del Giordano (FG) section (*Cr.* = *Crassicollaria*, *C.* = *Calpionella*, *Cp.* = *Calpionellopsis*, *Ct.* = *Calpionellites*, *T.* = *Tintinnopsella*, *R.* = *Remaniella*, *St.* = *Stomiosphera*, *Gl.* = *Globochaete*).

with Saccocoma Agassiz as the most representative genus.

From 6 to 131 m the limestone sequence contains a rich calpionellid fauna and the biostratigraphy of this interval has been worked out using the standard calpionellid zonation by Allemann et al. [13].

The first calpionellids represented by the genus *Crassicollaria* appear at 6 m followed by the genus *Calpionella* with the species *C. alpina* Lorenz at 12.5 m. As the zonal boundary between the *Crassicollaria* and *Calpionella* biozones cannot be drawn with confidence a combined *Crassicollaria / Calpionella* Zone (Late Tithonian/Early Berriasian) has been defined from 6 to 58.5 m. The upper boundary of this biozone coincides with the first appearence of the genus *Calpionellopsis.*

The *Calpionellopsis* Zone (Late Berriasian/Early Valanginian) starts at 58.5 m and continues upward to 105.5 m where the first occurrence of the species calpionellites darderi is recorded. The *Calpionellites* Zone (Valanginian) characterized by the presence of *C. darderi* has been located between 105.5 m and 131 m. Unfortunately, the upper boundary of this biozone which is marked by the disappearance of all calpionellids appears to coincide with the level where the second sampling gap between 131 m and 194 m starts. (Three samples collected from 130 m, 130.9 m and 131 m all show traces of Radiolaria but only the sample from 130.9 m contains a few specimens of *Calpionellopsis* and *Calpionellites*).

It is to be noted that the interval corresponding to the *Saccocoma* and the base of the *Crassicollaria / Calpionella* biozones is marked by a reduced thickness in which stratigraphic gaps may also be present. It appears, however, that the sampled sequence is biostratigraphically complete from above the first sampling gap between 18 m and 30 m to 131 m, where the next stratigraphic gap is indicated.

Higher up the sampled section only Radiolaria, Stomiosphera and Foraminifera are found and the interval between 194 m (131 m?) and 275.5 m has been named the Radiolaria Zone (Hauterivian).

The faunal assemblages recognized both in the Gubbio and Cagli sections are characteristic of the *Calpionella* Zone (Early Berriasian).

6. Magnetic stratigraphy of the Fonte del Giordano section

The virtual geomagnetic polar (VGP) latitudes for the Fonte del Giordano section (Fig. 7) were computed from the tectonically corrected palaeomagnetic sample (site) directions after thermal cleaning. (The VGP latitude is the geomag-

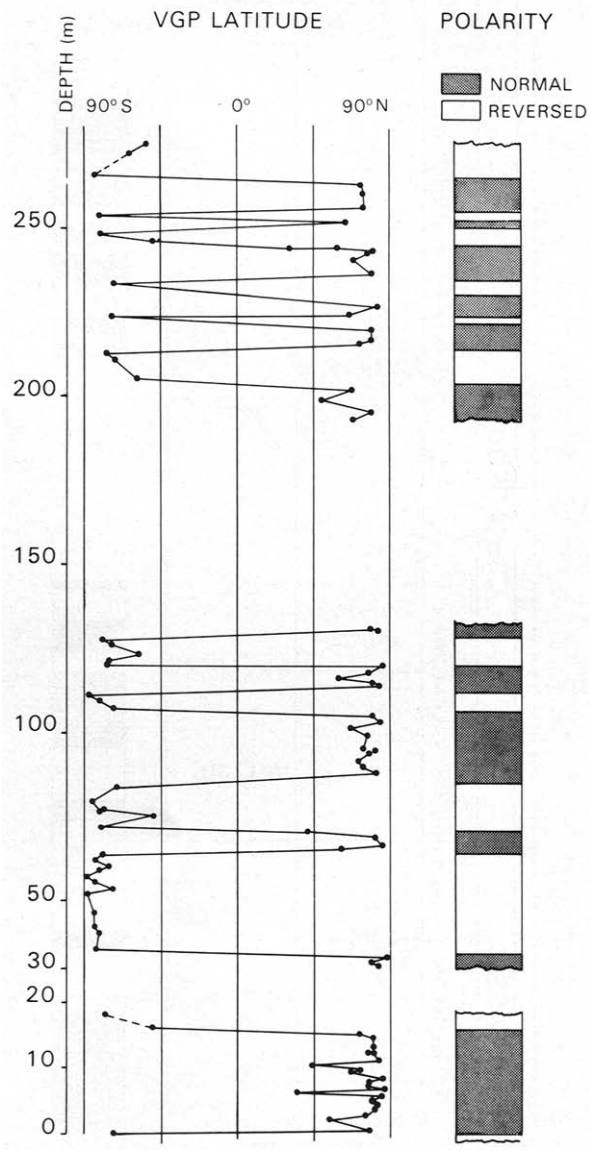


Fig. 7. VGP latitudes and magnetic polarity zonation in the Fonte del Giordano section.

netic latitude of the VGP for a sample in a centred dipole field defined by the mean VGP for all samples.) The polarity zonation has been derived from the VGP latitudes in Fig. 7 and the data provide a well defined magnetic stratigraphy except for two intervals that were not sampled. The boundaries between the normal and reversed polarity zones are precise within 0.7 m between 0 and 18 m, 2 m between 30 and 131 m and 2.7 m between 194 and 275.5 m, respectively.

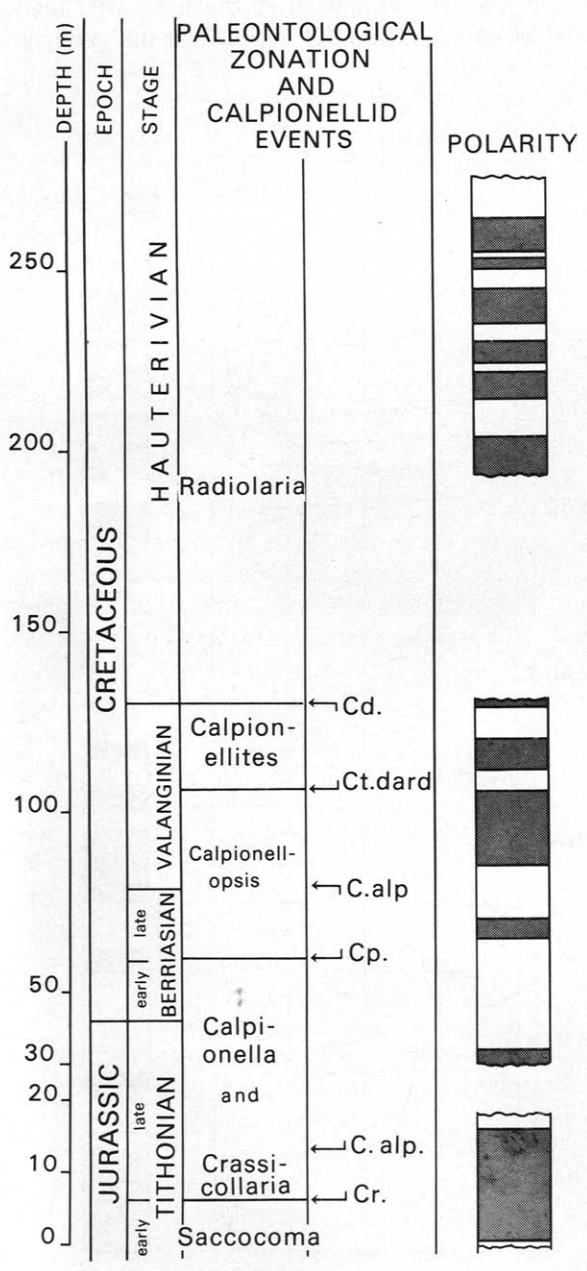


Fig. 8. Biostratigraphy and magnetic stratigraphy of the Maiolica limestones in the Fonte del Giordano section.

The magnetostratigraphic section (Fig. 8) is subdivided into three parts by the sampling gaps between 18 and 30 m and between 131 and 194 m, respectively.

For the lower subsection (0–18 m) the biostratigraphy has indicated a reduced thickness with occasional hiatuses. Therefore it is not entirely improbable that the normal polarity interval from 1 to 17 m is composed of more than one normal polarity zone with one or two unrecorded reversed zones in between. An approximate correlation to the M-sequence of oceanic magnetic anomalies [14] made on the basis of stratigraphical ages suggests that this normal polarity zone may correspond to the interval between anomaly M-18 and M-20 time (Fig. 9).

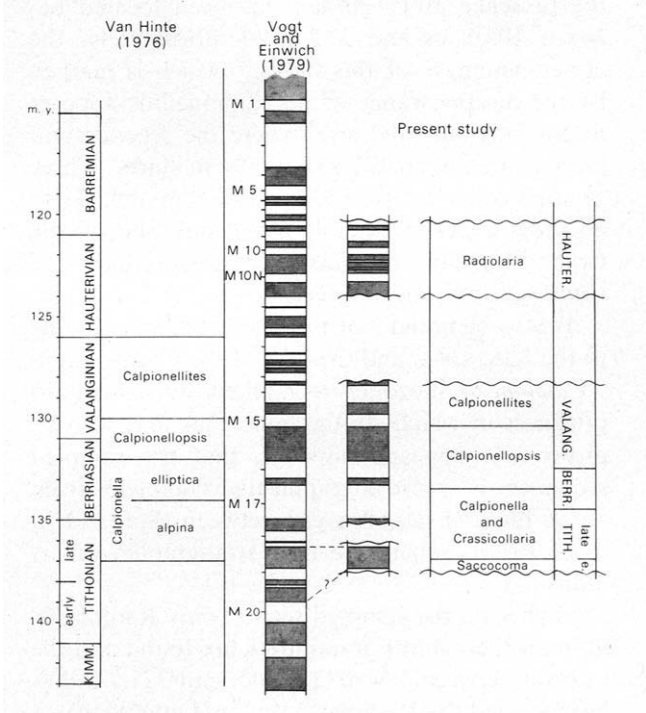


Fig. 9. Comparison of the biostratigraphy and magnetic reversals of the Fonte del Giordano section to the M-sequence of oceanic magnetic anomalies [14] and the standard calpionellid zonation [13] dated by Van Hinte [15].

The reversal sequence of the next subsection between 30 m and 131 m consists of well-defined alternating normal and reversed polarity zones that are easily identifiable with the M-sequence of oceanic magnetic anomalies between the normal polarity zones below M-17 and above M-14, respectively (Fig. 9). This correlation involves a sedimentation rate of 14 m/m.y. which is in good agreement with the average depositional rate of the Maiolica limestones in Umbria as suggested by their enormous thickness.

The upper subsection between 194 m and 275.5 m also displays alternating normal and reversed polarities including a few one sample reversals which, however, are based on more than one specimen each (Fig. 7). In assigning oceanic anomalies to these magnetozones we have made an approximately linear interpolation for the missing 63 m thickness below using the above derived 14 m/m.y. sedimentation rate and associated the lowermost reversed zone with anomaly M-10N. When each recorded polarity zone is correlated to a reversal zone of the M-sequence the topmost reversed polarity will be identified with anomaly M-8, giving an average depositional rate of 23 m/m.y. for this subsection. Though the latter accumulation rate is still realistic for the Maiolica the possibility that we have missed a few short reversals cannot be excluded entirely owing to the large (2.7 m) sampling distance adopted over this biostratigraphically poorly controlled subsection. So, it is not quite impossible that the uppermost reversed magnetozone is younger than anomaly M-8, but it is definitely not younger than anomaly M-3 the last long reversed zone before the Cretaceous quiet period preceded by frequent reversals.

7. Discussion and conclusions

Recently, Channel et al. [16] have reviewed recorded magnetic polarities during the Jurassic and Cretaceous both in land and sea.

The magnetostratigraphic results from the Fonte del Giordano sequence are directly comparable with those from the pelagic limestome sequence at Foza (Southern Alps) [17,18]. Starting deep in the Tithonian saccocomids and calpionellids give the zonation in both sections but the Foza section is discontinued within the *Calpionella elliptica* Subzone which may be parallelled conditionally to the interval between 46 m and 58.5 m in the Fonte del Giordano sequence. Channell et al. [16] report that in the Foza section "a fingerprint" pattern of magnetozones has been recognized that closely resembles the oceanic polarity pattern between anomalies M-16 and M-22 (Fig. 9). The broad reversed magnetozone in this "fingerprint" is correlated with anomaly M-17 similarly to what has happened in the present work.

The magnetostratigraphic Cismon section [19] covers a period from the Valanginian to the Campanian. Calpionellids are only present in the lowermost part and can all be referred to the *Calpionellites* Zone. Channell et al. [16,19] have correlated the Valanginian/Hauterivian part of the section to the oceanic reversal pattern approximately between anomaly M-11 and M-3. The *Calpionellites* Zones in the Cismon and Fonte del Giordano sections do not overlap owing mainly to the sampling (and perhaps stratigraphic) gap at 131 m in the latter series, but the Hauterivian segment of the Fonte del Giordano polarity record is in agreement with the corresponding Cismon reversal sequence in that both are showing mixed polarities.

The magnetic reversal sequence obtained from the white pelagic limestones of Late Jurassic/Early Cretaceous age at Sümeg (Transdanubian Central Mountains) [20] is correlated to the oceanic polarity time scale using saccocomids and calpionellids from three distinct intervals for more precise age definition. The long reversed zone near the Jurassic/Cretaceous boundary which is interrupted only by two normal samples is identified with anomalies M-16 and M-17 while above and below this feature normal polarities prevail. The correlation we have suggested for the corresponding part of the Fonte del Giordano reversal sequence is practically the same as that for the Sümeg sequence.

In conclusion, we note a broad agreement of the interpreted magnetic reversal sequence of the Fonte del Giordano section (Fig. 9) with corresponding segments of magnetic stratigraphies from land sections in the peri-Adriatic region.

It appears that the Fonte del Giordano section from 0–131 m is the first sequence which comprises the complete calpionellid zones and has been studied magnetostratigraphically as well. We admit that the presence of a break from 18 to 30 m may introduce ambiguity into the magnetostratigraphic correlation but this only applies to the lower 30 m of the section. From 30 to 131 m an unequivocal magnetic reversal pattern was recorded allowing a one-to-one correlation both of magnetozones and calpionellid events to oceanic magnetic anomalies (Fig. 9). The extrapolation we have made for the missing 131–194 m of the section is obvious and the presence of mixed polarities in the Radiolaria Zone from 194 to 275.5 m is indicative of the top of section being definitely older than anomaly M-3 time.

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